

CO 2 - ASSESSMENT TOOL -2

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Hotel Reservation and Room Management Platform with Dynamic Pricing

Analyze System Requirements

System Objectives

- Enable online hotel room reservations.
- Manage room availability in real time.
- Implement dynamic pricing based on demand, occupancy, and season.
- Provide secure online payment.
- Generate booking and revenue reports.
- Improve customer satisfaction and hotel revenue.

Functional Requirements

- User Registration and Login
- Search Available Rooms
- View Room Details
- Book Room
- Cancel Booking
- Online Payment
- Dynamic Pricing
- Room Management
- Booking Management
- Customer Management
- Reports and Analytics
- Notifications

Non-Functional Requirements

- High Performance
- Security
- Reliability
- Scalability
- Availability
- Maintainability
- Usability

Quality Attributes

- Scalability
- Maintainability

- Reliability
- Availability
- Performance
- Security

2. Select Suitable Software Architecture

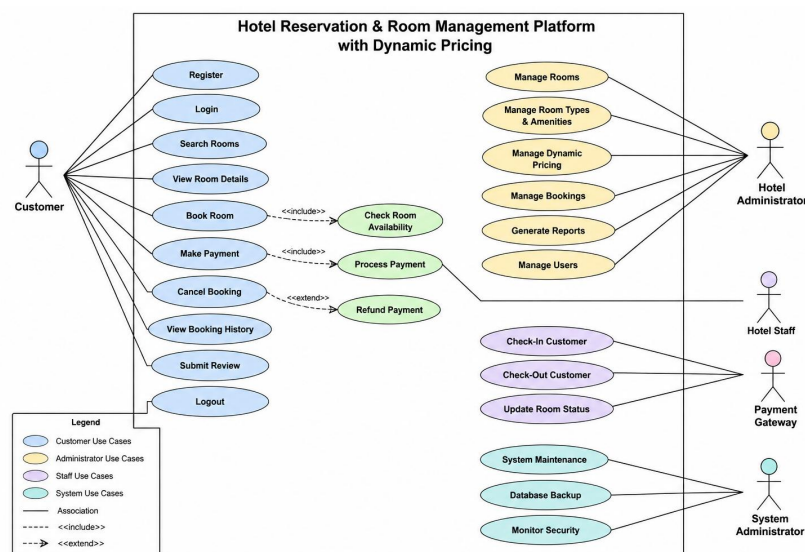
Selected Architecture

Layered Architecture (Three-Tier Architecture)

Justification

- Separates presentation, business logic, and database layers.
- Easy to maintain.
- Highly scalable.
- Better security.
- Easier testing.
- Supports future enhancements.

3. Software Architecture Diagram



4. Components and Their Interactions

Presentation Layer

Responsibilities:

- User Interface
- Login

- Search Rooms
- Booking
- Payment Screens

Business Logic Layer

1. Authentication Service
2. Booking Service
3. Dynamic Pricing Engine
4. Payment Service
5. Notification Service
6. Report Service

Data Access Layer

Database

Stores:

- Customer Information
- Room Details
- Booking Records
- Payment Records
- Pricing Rules
- Reports

System Workflow

1. User logs into the system.
2. Searches available rooms.
3. Booking service checks room availability.
4. Dynamic Pricing Engine calculates the room price.
5. User confirms booking.
6. Payment Service processes the payment through the Payment Gateway.
7. Booking is stored in the database.
8. Notification Service sends confirmation via Email/SMS.
9. Administrator manages bookings and generates reports.

5. Quality Attribute Discussion

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability

6. Architectural Justification

Reasons for Choosing Layered Architecture

- Clear separation of responsibilities.
- Easier debugging and maintenance.
- Supports future expansion.
- Enhances code reusability.
- Simplifies testing.
- Improves security.

Trade-offs

Advantages

- Easy maintenance
- High scalability
- Better security
- Modular design
- Easy integration

Disadvantages

- Slightly slower due to multiple layers.
- More initial development effort.

Future Enhancements

- AI-based room recommendation.
- Machine learning for demand prediction.
- Mobile application.
- Multi-language support.